



# Investigation report

## Rejected take-off and runway overrun, Tarlton Springs - Causal and contributory factors report

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INVESTIGATION WORKFLOW SUITE

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## Synopsis

On 18 June 2026, Saltbush Air SB512, a 50-seat twin-turboprop registered VH-SBQ, was taking off from runway 09 at Tarlton Springs Aerodrome at about 17:30 local time, five minutes after the end of civil twilight, with 44 passengers, two flight crew and one cabin crew member on board.

Two seconds after V1, the first officer saw two kangaroos crossing the runway ahead. At 07:30:32 UTC the commander rejected the take-off at 118 kt, six knots above the calculated V1 of 112 kt. About one second later the No. 2 propeller and right main landing gear struck both animals, destroying the No. 2 propeller and leaving reverse available on No. 1 only. Maximum braking was established two seconds after the decision, faster than the fleet average, but the aircraft left the runway end at approximately 21 kt. It stopped upright on the extended centreline, 62 m into the 90 m runway end safety area.

The aircraft was substantially damaged: three No. 2 propeller blades were damaged beyond limits, the right main gear doors were distorted, both right main tyres deflated through their fuse plugs, and the lower right fuselage was punctured. There were no serious injuries. One cabin crew member reported bruising during disembarkation by the forward airstair, which was made without an evacuation. Two kangaroos were killed, and the aerodrome was closed for 4 hours 20 minutes.

The kangaroos' presence on the sealed surface extremely likely led to both the reject and the strike. The decision to reject above V1 extremely likely contributed, because the scheduled accelerate-stop distance assumed a stop begun at or before V1 and left a margin of only 67 m. It is virtually certain that the loss of No. 2 reverse left deceleration approximately 15% below the certified value, and virtually certain that the late start and reduced deceleration together carried the aircraft beyond the runway end, neither alone being sufficient.

The contributory factors were:

- **Northern perimeter fence:** repair was deferred, unfunded, at three budget rounds since 2023, which very likely provided the animals' route onto the runway.
- **Runway inspection requirement:** an inspection before the first departure after last light was required only at aerodromes listed after a reported strike, so Tarlton Springs could not be listed despite eleven recorded sightings of kangaroos on the sealed surface. This very likely contributed.
- **Inspection not requested:** nothing prompted either pilot to consider one, which likely contributed.
- **Timing:** the departure fell within the period of peak recorded kangaroo movement, which likely contributed.
- **Evening service approval:** the service was approved as a schedule change without reopening a route risk assessment restricting the aerodrome to daylight. This likely contributed, although a review might have been confined to lighting and fuel.
- **Rejected take-off training:** the training addressed only aircraft system failures. This likely contributed, although the commander's account supports a deliberate departure from the rule.
- **Aerodrome standard threshold:** a documented wildlife plan was required only above a size threshold the aerodrome did not meet. This likely contributed, with less confidence than the other factors.

The crew's long duty was unlikely to have influenced the occurrence but remains a standing risk. A partially dark cabin floor proximity escape path, undetectable by routine checks, did not influence the occurrence but remains a safety issue. A tailwind and a No. 2 engine power loss were both very

unlikely. Whether runway edge lighting intensity affected detection of the animals could not be resolved.

After the occurrence the operator suspended the evening Tarlton Springs service. No other safety action has been recorded. [NOT ESTABLISHED: whether any safety recommendation is made]

## Factual information

### History of the flight

On 18 June 2026, Saltbush Air SB512 was the last scheduled sector of the day from Tarlton Springs Aerodrome to Sydney and the fifth sector of the crew's duty, added that morning when the rostered crew reported unfit. The doors closed at 07:16, 40 minutes behind schedule, with 44 passengers, two flight crew and one cabin crew member on board. The aerodrome was in darkness; the tower was open and rostered to close at 18:00 local. The first officer was pilot flying.

At 07:29:10 the tower cleared the aircraft for take-off on runway 09, passing the wind as 090° at 6 kt. Neither the tower nor the crew raised a runway inspection, and no transmission about wildlife was made on the tower channel in the preceding 90 minutes. The only runway inspection that day had been at 21:35 on 17 June (UTC), before the first scheduled arrival, and had recorded the movement area as clear.

Take-off power was set at 07:30:02 and the commander called V1 at 07:30:29. Two seconds later the first officer saw two kangaroos crossing the runway from the northern side, beyond the 1,100 m point, at an estimated range of around 200 m, which she later said could have been less. At 07:30:32, at 118 kt, the commander took control and rejected the take-off, six knots and three seconds past the calculated V1 of 112 kt, with 695 m of sealed runway remaining. About one second later the aircraft struck both animals with the right main landing gear and the No. 2 propeller.

Maximum braking was established at 07:30:34. The commander selected reverse at 07:30:36, but only the No. 1 propeller responded. The aircraft left the end of runway 09 at 07:30:51 at approximately 21 kt and stopped at 07:30:57, 62 m into the 90 m runway end safety area (RESA), upright on the extended centreline with the nose wheel in graded gravel, 28 m short of the end of the prepared area, beyond which the terrain is ungraded.

There were no fire indications. The commander set the parking brake, shut down both engines and told the cabin to remain seated. At 07:31:54 he told the tower that the aircraft had gone off the end of runway 09, with no injuries and no fire, and requested activation of the emergency plan, which the tower activated at 07:32:10, dispatching a vehicle. No evacuation was commanded and no slides were deployed. Passengers disembarked by the forward airstair with the main cabin lights on, attended by aerodrome staff. The occurrence notification states that passengers were disembarked 4 minutes 8 seconds after the aircraft stopped; the timeline places disembarkation between 07:34:40 and 07:38:48. [NOT ESTABLISHED: time at which the last passenger was clear of the aircraft]

The aerodrome was closed for 4 hours 20 minutes. Both flight crew were stood down and tested for drugs and alcohol, with negative results. The operator's Head of Flying Operations suspended the evening Tarlton Springs service.

### Recorded information

The FDR and CVR were removed by the operator's engineering representative at 12:40 on 18 June and sent to the operator's flight data section. FDR and QAR data showed both engines at take-off torque, 98.4% and 98.5%, from power set until the power levers were retarded at 07:30:32, with no exceedance, caution or warning before then, and fuel flow, ITT and propeller RPM within normal scatter. The headwind component was 4 to 6 kt throughout the roll.

- **Reject:** 118 kt IAS and 115 kt groundspeed; both power levers at flight idle within 0.6 seconds.

- **Impact:** at 07:30:33, 114 kt, with spikes of 0.31 g lateral to the right and 1.48 g vertical.
- **Maximum braking:** 2,940 psi at 07:30:34, two seconds after the decision, with anti-skid active on all four wheels, against an operator fleet mean of 3.2 seconds.
- **Reverse:** only No. 1 reverse was selected. No. 2 propeller RPM decayed from 07:30:33 and the No. 2 reverse interlock did not make.
- **Deceleration:** a mean of 2.6 m/s<sup>2</sup> between 07:30:34 and 07:30:44, approximately 15% below the certified accelerate-stop value for the weight and runway condition, which assumes symmetric reverse; 1.9 m/s<sup>2</sup> in the gravel.

Distance reconstruction placed the aircraft 1,152 m along runway 09 at the reject. The stopping distance achieved from 118 kt was 757 m.

The CVR recorded the first officer at 07:30:31: "Roos - roos, roos, on the left, they're on the -". The commander replied at 07:30:32: "Stopping. I have control." At 07:30:36: "Reverse - number two's not coming. Max brakes, max brakes." At 07:30:47: "We're not stopping on it."

After the aircraft stopped, the first officer said the animals "came out of the dark on the left of the strip, crossed the centreline". The commander said: "I never had them. I was heads-in on the engine page after V one." In his written safety report, however, he stated that he looked up after the first officer's call and saw the two animals on the sealed surface ahead before rejecting.

## Aircraft information

VH-SBQ is a 50-seat twin-turboprop, at 20,800 kg with flap 10 for this departure. The calculated V1 assumed a dry runway 09 and a 5 kt headwind component. [NOT ESTABLISHED: aircraft type designation, maintenance history and pre-flight serviceability status]

- **No. 2 propeller:** three blades damaged beyond repairable limits, with leading-edge deformation and one blade tip separation.
- **Right main landing gear:** distorted gear doors; both tyres deflated through the fuse plugs at a calculated brake energy of 84% of the certified maximum.
- **Lower right fuselage:** abrasion and two punctures consistent with propeller debris.

## Meteorology

The wind passed by the tower was consistent with the 07:00 routine observation and the recorded headwind component. The runway was dry.

## Personnel information

**Commander.** 8,410 hours total, 2,140 on type, in command for 3 years 1 month, with one previous turboprop type rating. The crew duty record gives his last cyclic simulator check as 11 March 2026, all items satisfactory; at interview it was given as a recurrent simulator check in February 2026. His last line check, on 2 December 2025, was satisfactory. He had no prior occurrence history.

**First officer.** 1,655 hours total, 405 on type. Her last cyclic simulator check, on 28 April 2026, was satisfactory in all items. She had flown two previous sectors into Tarlton Springs, on 9 January and 17 February 2026, both in daylight.

**Duty.** The crew duty record gives sign-on as 06:30 local, placing the occurrence 11 hours 00 minutes into a planned duty of 12 hours 05 minutes, against a maximum of 12 hours 30 minutes for five sectors with that sign-on. At interview the commander's duty was recorded as beginning at 05:10 local. Duty and rest over the preceding seven days were within company limits, with no variations applied. Both pilots described themselves as tired but not impaired in their decision-making.

## Aerodrome information

The 90 m RESA is graded gravel. The post-occurrence survey found two carcasses at 1,188 m and 1,221 m from the threshold, continuous tyre marks from 1,160 m to the runway end, and a nose wheel rut extending 62 m into the RESA on the extended centreline.

The commander described the runway lighting as a corridor of light with darkness either side, but could not say whether this affected his view. The first officer described it as normal, her only comparison being two daylight landings. [NOT ESTABLISHED: runway edge lighting intensity setting at the time of the occurrence]

[NOT ESTABLISHED: condition and repair history of the aerodrome perimeter fence]

## Organisational and management information

Under the operator's procedures (Part A 8.14 and the Appendix 8A list), a runway inspection is requested before the first departure after dark at listed aerodromes, the listing criterion being a previous wildlife strike. Tarlton Springs was not listed. [NOT ESTABLISHED: text of Part A 8.14 and Appendix 8A, and the outcome of the Safety Department's review of 3 March 2025]

The evening service had been introduced as a schedule change; the commander said no briefing about it reached him. [NOT ESTABLISHED: approval process for the evening service and whether the route risk assessment was reviewed]

Before the flight the first officer consulted the flight plan, charts, the ERSA entry and the company aerodrome brief page; none mentioned animals.

The commander stated that all the stop-or-go scenarios in his training were engine failures below V1. [NOT ESTABLISHED: content of the operator's rejected take-off training syllabus]

[NOT ESTABLISHED: the aerodrome's wildlife management plan obligations and the size threshold applied]

## Tests and research

Shock-load inspection and borescope examination of both engines found no pre-existing damage and no evidence of any power anomaly before the impact. Animal tissue was recovered from two No. 2 propeller blade roots. The right main gear structure was within limits, there was no brake fire or hydraulic leak, and the anti-skid system tested serviceable.

During disembarkation the first officer noticed unlit sections of the cabin floor proximity strip aft, with the emergency lighting armed and the ceiling lights on. A functional check was carried out on 24 June 2026. [NOT ESTABLISHED: result of the floor proximity lighting functional check]

## Other information

The Tarlton Springs wildlife sighting log for January 2025 to June 2026 contained 46 entries: 41 recorded kangaroos on or adjacent to the movement area, 38 of them between one hour before and two hours after last light, and 11 recorded animals on the sealed surface of runway 09/27. The log is a paper register held in the aerodrome office and is not reported to any operator; the aerodrome reporting officer confirmed that no airline had ever requested it. The commander stated that he had seen kangaroos on the runway strip at Tarlton Springs in the evening on several occasions, but had not submitted a report.

Two kangaroos were killed. One cabin crew member reported bruising during disembarkation; there were no serious injuries.

## Analysis

### External Influences

#### **Documented wildlife plan required only above a size threshold the aerodrome does not meet**

The aerodrome's wildlife arrangements were shaped by a regulatory threshold that it did not meet. Aerodrome Standard AS-139 section 10.4 sets out the obligation in three parts:

> 10.4.1 The operator of a certified aerodrome must take reasonable measures to reduce the risk to aircraft operations presented by wildlife on or in the vicinity of the aerodrome.

>

> 10.4.2 The operator of a certified aerodrome must prepare, implement and maintain a documented wildlife hazard management plan where the aerodrome is used by aeroplanes having a maximum take-off weight exceeding 27,000 kg, or serves more than 350,000 passenger movements in a calendar year.

>

> 10.4.3 Where subsection 10.4.2 does not apply, the measures taken under 10.4.1 need not be documented, but the aerodrome operator must be able to demonstrate what measures are in place.

Tarlton Springs is a certified aerodrome. Its heaviest scheduled aeroplane is the 22,900 kg twin-turboprop used on the Saltbush Air service, and it handled 61,400 passenger movements in 2025. Neither limb of 10.4.2 therefore applied. Under 10.4.3 the aerodrome operator was not required to document its measures, and it held no wildlife hazard management plan, no hazard assessment, no dispersal programme and no record of measures beyond the single pre-first-arrival inspection at Aerodrome Manual 6.3. This position was confirmed against the certified aerodrome register and against the aerodrome operator's own compliance file. These facts are documentary and not in dispute, and it is virtually certain that the condition existed.

The aerodrome operator was not in breach of the standard. The matter therefore concerns the design of the standard rather than compliance with it. At an aerodrome of this size, the threshold left a wildlife programme that was undocumented, unaudited and, in practice, unfunded.

The question for the analysis is whether this condition influenced the occurrence. The evidence points both ways.

#### **The case for influence**

Two items of information about the hazard existed at the aerodrome but did not reach the people who could act on them.

- **The sighting log.** Between January 2025 and June 2026 the aerodrome's wildlife sighting log recorded 46 entries. Of these, 41 recorded kangaroos on or adjacent to the movement area. Of those 41, 38 were timed between one hour before and two hours after last light, and eleven recorded animals on the sealed surface of runway 09/27. The log was a paper register kept in the aerodrome office. It was not reported to any operator, and the aerodrome reporting officer confirmed that no airline had ever asked for it.
- **The works register.** The northern perimeter fence between the 700 m and 1,300 m runway markers was recorded in item WR-221 in April 2023 as failed, with animals observed passing through it. Replacement was estimated at \$41,000. The item was

deferred as unfunded at the 2023, 2024 and 2025 budget rounds. Item WR-263, raised in February 2026, recorded the same section as deteriorating further. A site inspection on 23 June 2026 found a continuous 140 m section with the bottom two wires separated from the posts and an animal pad through it at the 1,050 m mark.

A documented plan is the one mechanism that would have linked the sighting log to the operators using the aerodrome. It is also the mechanism that would have placed the failed fence on a funded programme rather than an unfunded works register. A documented plan brings an audit trail and an owner. Each of the other controls examined in this occurrence failed for want of exactly those things.

### **The case against influence**

The absence of a documented plan did not remove the aerodrome operator's underlying duty. Section 10.4.1 still required it to take reasonable measures. A documented plan is not the only way to repair a fence or to pass a sighting log to an airline. The deferral of the fence repair was a budget decision by the council. There is no evidence that a documented plan would have secured funding where the works register did not. A plan is a document, whereas the fence repair required money.

### **Resolution**

The investigation weighed these arguments and accepted, on the balance of the evidence, that the absence of a documented plan influenced the occurrence. The conclusion rests on two points:

- The sighting log and the fence defect were both known at the aerodrome.
- Neither was connected to any decision by the operator or by those allocating funds, and the missing link was the one a documented plan exists to provide.

The contrary argument has force and is not dismissed. It is possible that a documented plan would have existed on paper while the fence remained unfunded. For that reason the conclusion is held with less confidence than the other contributing factors in this occurrence, and it should not be read as settled.

The condition acted through two intermediate factors, each analysed in its own subsection:

- the deferral of the perimeter fence repair over three budget cycles; and
- the operator's arrangement under which runway inspections were required only at listed aerodromes, with listing triggered by a strike.

Its effect was therefore indirect. Removing it would not by itself have prevented the occurrence, because the fence could have remained unrepaired and the inspection criterion unchanged even with a plan in place. The factor is contributory rather than causal.

It is likely that the absence of any requirement for Tarlton Springs to hold a documented wildlife hazard management plan contributed to the occurrence. It did so by leaving the aerodrome's knowledge of kangaroo activity and of the failed fence without an owner or a route to the operator. This conclusion is held with less confidence than the other contributing factors.

## **Organisational Influences**

### **Perimeter fence repair deferred for three budget cycles**

The aerodrome operator's works register shows that the northern perimeter fence was known to have failed more than three years before the occurrence. Item WR-221, raised in 04/2023, described four displaced panels and strained wire on the northern boundary between the 700 m and 1,300 m runway markers. It noted that animals had been observed passing through, and recommended replacement at an estimated cost of \$41,000. The item was deferred at the 2023, 2024 and 2025 budget rounds. On each occasion the entry recorded it as unfunded and carried it forward for

review at the next cycle. Item WR-263, raised in 02/2026, recorded that the same section was deteriorating further.

The site inspection on 23/06/2026 found the fence in the condition the register described. A continuous 140 m section of the northern fence had its bottom two wires separated from the posts, and a clear animal pad ran through it at the 1,050 m mark. Because the aerodrome operator's own documents and the physical inspection agree, it is virtually certain that the fence repair was deferred and that the breach existed on the day of the occurrence. Nothing about the breach had to be reconstructed. The aerodrome's knowledge of it, and of animals using it, was set down in its own register from the first entry.

The link between the breach and the occurrence rests on position. The first officer saw the animals come onto the runway from the northern side, beyond the 1,100 m point. The two carcasses were recovered at 1,188 m and 1,221 m from the runway 09 threshold. These are on the same side of the runway as the breach and a short distance downstream of the animal pad at 1,050 m. A fence in the condition found does not exclude kangaroos. The aerodrome's wildlife sighting log for 01/2025 to 06/2026 shows how often they reached the movement area: 41 of its 46 entries recorded kangaroos on or adjacent to the movement area, and eleven recorded animals on the sealed surface of runway 09/27. It is very likely that the deferred repair influenced the occurrence, by leaving open the route through which the two animals reached the runway strip.

The evidence identifies the breach as the very likely route. It does not establish that an intact northern fence would, on its own, have kept the animals off the runway that evening. [NOT ESTABLISHED: the condition of the remainder of the aerodrome perimeter, and whether any other entry route was available to the animals.] The deferral is therefore treated as contributory rather than causal. Why the aerodrome's wildlife arrangements did not require the breach to be treated as a hazard is addressed under External Influences. How the animals came to be on the sealed surface at the moment of the take-off roll is argued with the occurrence events.

The perimeter fence repair was deferred for three budget cycles, and this very likely contributed to the occurrence by leaving open the route through which the animals reached the runway.

### **Evening service approved as a schedule change, route risk assessment not reopened**

The operator's current route risk assessment for Tarlton Springs is RRA-041 issue 3, dated 11/2022. It assessed hazards including "runway length and accelerate-stop margin" and "single runway with no lighting redundancy". It carried an operating restriction of "daylight operations only pending review of night lighting and diversion fuel policy". No later issue exists.

Change Management Record CMR-2025-118 introduced "one evening rotation, Sydney to Tarlton Springs to Sydney, six days per week, from 14/07/2025". This was the first scheduled company operation into Tarlton Springs after last light. The change was assessed as a "schedule change only", with "No new hazard identified", and the question whether a route risk assessment review was required was answered "No". The Head of Flying Operations approved it on 22/05/2025. The daylight-only restriction is not referenced anywhere in CMR-2025-118, and RRA-041 was not reopened. The Safety Department's records show no involvement in the change beyond receipt of the approved document.

Both documents are current controlled copies, so it is extremely likely that the change was approved in this way without the route risk assessment being reopened. The approver, who has since left the company, was not interviewed. As a result, the reasoning behind the assessment is known only from the document itself. The process asked whether the change was new. It did not

ask whether the existing assessment still held for the operation now proposed. That was how a live restriction against night operations was passed over.

The question for influence is what a review of RRA-041 against a night operation would have reached. Such a review would have had to consider the aerodrome after last light. That consideration would have led to the aerodrome's wildlife exposure and to whether Tarlton Springs should appear in Appendix 8A of the Operations Manual. Under paragraph 8.14.2, a listed aerodrome requires the pilot in command to "request a runway inspection from the aerodrome operator prior to the first departure after last light". The argument against is that the restriction names only night lighting and diversion fuel, and a review might have addressed those two matters alone without reaching wildlife. For this reason the influence is put as likely rather than higher.

The pathway is indirect. The change did not itself put the animals on the runway. It removed an opportunity for the operator to recognise the exposure and to bring its inspection control to bear. The factor is therefore contributory. The design of the inspection control and its listing criterion are argued under Risk Controls, and the absence of an inspection on the evening of the occurrence is argued under Individual Actions.

The evening service was extremely likely approved as a schedule change without the route risk assessment being reopened, and this likely contributed to the occurrence by removing the point at which the aerodrome's night-time wildlife exposure could have been identified and controlled.

## Risk Controls

### **Runway inspection required only at listed aerodromes, and the listing criterion is a strike**

The operator had a defence written for this kind of occurrence. Operations Manual Part A 8.14.2 states that "At an aerodrome listed in Appendix 8A, the pilot in command shall request a runway inspection from the aerodrome operator prior to the first departure after last light, and shall not commence the take-off roll until the result of that inspection has been passed." At any other aerodrome, 8.14.3 provides that "no specific action is required."

Appendix 8A admits an aerodrome only "where a wildlife strike involving a company aircraft has been reported at that aerodrome within the preceding 24 months." Nine aerodromes were listed on 18/06/2026. Tarlton Springs was not among them and never had been. No provision in Part A asks a crew to report a sighting, and none admits an aerodrome to the list on any other evidence. The list was last reviewed on 03/03/2025, four months before the evening service began.

The hazard at Tarlton Springs was already documented locally. The aerodrome's wildlife sighting log for 01/2025 to 06/2026 contains 46 entries:

- 41 record kangaroos on or next to the movement area.
- 38 of those 41 are timed between one hour before and two hours after last light.
- 11 record animals on the sealed surface of runway 09/27.

The log is a paper register held in the aerodrome office. It is not reported to any operator, and the aerodrome reporting officer told the investigation that no airline had ever asked for it. None of these sightings could bring the aerodrome onto the list, because none was a strike involving a company aircraft.

The control was therefore unarmed at the aerodrome where it was most needed. Its trigger was circular: it could only be activated by the event it existed to prevent. It is virtually certain that this condition existed, because both the manual text and the appendix listing are documentary.

It is very likely that the condition influenced the occurrence. Had 8.14.2 applied, an inspection would have been requested and its result passed before the roll, and the aerodrome's staff disperse animals when they find them. The investigation did not put the influence higher than very likely, because an inspection completed some minutes before departure does not guarantee a clear runway at the moment of the roll. The same point settles how this factor is classified. The animals entered through a perimeter known to be failing, and they entered at the peak of kangaroo movement. An inspection would very likely have changed the outcome, but it cannot be said that the occurrence would not have happened without this gap.

The two upstream conditions are argued elsewhere: the documented wildlife plan requirement that did not apply at Tarlton Springs, and the approval of the evening service as a schedule change. The absence of a runway inspection on the night is argued under Individual Actions.

The inspection requirement's dependence on a reported strike was a contributory factor in the occurrence, and it is a safety issue in its own right.

### **Rejected take-off training covers aircraft system failures below V1 only**

Recurrent training syllabus TRG-SYL-07, module 4, requires crews to demonstrate a rejected take-off "for an engine failure, an engine fire and a configuration warning, each initiated below V1", and the decision to continue for a failure at or after V1. Its four scripted scenarios are:

- engine failure at 90 kt;
- engine fire at 100 kt;
- configuration warning at 85 kt;
- engine failure at V1 plus 5 kt, continued.

Every scenario is an aircraft system event. In the five training cycles held on file, no crew had been presented with an obstruction on the runway, and 214 crew completed the module in the 2025 to 2026 cycle alone.

Part B 4.6 is consistent with the syllabus. It states that "At or above V1 the take-off shall be continued", and it contains no guidance on an obstruction seen on the runway at any speed. Both flight crew independently described their stop or go training in the same terms, without being prompted. It is extremely likely that this gap existed. The figure stops short of certainty only because scenario data older than five cycles was not available.

The influence of the gap on the decision was argued in both directions, and both arguments are set out here.

- **For influence:** the captain had about two seconds to decide. The decision he made at 118 kt was the first time he had considered the question at all. Preparing crews for that moment is what training is meant to achieve.
- **Against influence:** the captain stated that he knew the rule and departed from it deliberately, judging that continuing was the worse option. Training on obstructions would not have changed what he knew. On this reading, the gap is a finding about the fleet rather than a cause of this reject.

The investigation concluded that the gap likely influenced the decision. An unrehearsed decision made cold in about two seconds is not the same as one made against a briefed and practised frame. This conclusion explains the conditions under which the decision was taken; it does not excuse or judge the decision. The reject itself is argued under Individual Actions.

The absence of obstruction scenarios in rejected take-off training and guidance was likely a contributory factor in the take-off being rejected above V1.

## Cabin floor proximity emergency lighting partially unserviceable

The first officer noticed that sections of the aisle floor strip were unlit while walking back through the cabin during the disembarkation. A functional check on 24/06/2026 was carried out with the system armed and the aircraft on battery power. It found two of the six floor proximity escape path segments dark, on the left side aft of row 16. The fault was traced to a failed ballast common to both segments. It is virtually certain that this condition existed.

It is unlikely that the condition influenced the occurrence or its outcome. No evacuation was commanded, the main cabin lights were on throughout, and passengers left by the forward airstair without urgency 4 minutes 8 seconds after the aircraft stopped.

The condition is nevertheless very likely to be important to future safety:

- **The defect could not be detected in service.** The pre-flight emergency lighting test on this type illuminates the ceiling lights and exit signs but neither illuminates nor monitors the floor segments. Neither the crew nor a line engineer could have found the fault without a dedicated functional test.
- **The monitoring interval cannot catch it.** The last such test was 11 months before the occurrence, within the maintenance programme's 24-month interval. The true state of the floor lighting across the fleet is therefore unknown.
- **The component is common to the type.** The same ballast is fitted across the type.
- **The consequences could be serious in other circumstances.** In a night evacuation with smoke, or with passengers leaving over the wing, two dark segments in the aft cabin could decide whether the exit is orderly or blocked.

The partially unserviceable floor proximity lighting, and the monitoring regime that could not detect it, did not contribute to the occurrence and are a safety issue.

## Local Conditions

### Departure in the hour after last light, the peak of kangaroo movement

The take-off roll began at 17:30 local time. This was 32 minutes after sunset at 16:58 and five minutes after the end of civil twilight at 17:25. Moonrise was not until 22:41, and the moon was a waning crescent 14% illuminated. The aerodrome was in full darkness, with no natural ambient light on the strip and no town lighting within sight. The investigator visited the site at the same time of day on 23/06/2026 and confirmed that the unlit graded strip on either side of the sealed surface cannot be seen from the centreline.

The aerodrome's wildlife sighting log for the period from January 2025 to June 2026 contained 46 entries. Forty-one of these recorded kangaroos on or next to the movement area, and 38 of those 41 fell between one hour before and two hours after last light. Eleven entries recorded animals on the sealed surface of runway 09/27. The log is a paper register, and its completeness could not be verified. For that reason the pattern it shows was assessed as extremely likely rather than certain.

The first officer's call, "Roos - roos, roos, on the left", came at 07:30:31 UTC. By then the animals were already on the lit surface, crossing from the northern side at a range she estimated at about 200 m. Her account at 07:32:35 was that they "came out of the dark on the left of the strip". The animals were therefore not visible while they were on the unlit strip, which is a direct consequence of the light conditions. The departure also fell inside the period in which the aerodrome's own records show almost all of its kangaroo movement.

On its own this condition was unremarkable, because scheduled services operate after dark routinely. Its significance lies in combining a period of peak animal movement with darkness that

hid everything beyond the sealed surface. That combination is what made the absence of a runway inspection matter. The reasons no inspection was requested are analysed under Risk Controls and Individual Actions. The condition did not by itself place the animals on the runway, and its absence alone would not necessarily have prevented the occurrence.

It is likely that the departure in the hour after last light, during the aerodrome's peak period of kangaroo movement and with no ambient light on the strip, was a contributory factor in the animals reaching the sealed surface unseen.

### **Runway edge lighting intensity**

The investigation examined whether the intensity of the runway edge lighting reduced the crew's ability to detect the animals. The matter could not be resolved.

**Evidence supporting the possibility.** In interview, the commander described the lighting as brighter than he would have wished. He said it formed a corridor of light with everything outside it black, so that anything approaching from beyond the seal could not be seen. On 25/06/2026 a photometric check was carried out at the investigator's request. It found four edge lights on the northern side, between the 900 m and 1,200 m markers, operating at 118% to 131% of nominal intensity for stage 3. This is the section of runway the animals crossed: the carcasses were recovered at 1,188 m and 1,221 m from the runway 09 threshold. The remaining lights were within tolerance. The aerodrome operator's maintenance programme checks that lamps are serviceable but does not measure their intensity.

**Evidence against it.** The first officer was the crew member who saw the animals. She described the lighting as normal and said she saw them clearly once they were on the sealed surface. She also acknowledged that, with only two previous daylight sectors into Tarlton Springs for comparison, she was not well placed to judge whether the lighting was unusual. The commander's failure to see the animals has an explanation that does not depend on the lighting. At 07:32:47 he said: "I never had them. I was heads-in on the engine page after V one." This is where the procedure directs his attention after V1.

**Why it could not be resolved.** The measured deviation was real and lay in the relevant section of runway. However, the only measurement was taken seven days after the occurrence, and nothing establishes how the lights were operating on the night. Nor did any evidence establish a link between edge lights at up to 131% of nominal and the detection of an object on the runway. The two crew accounts were each credible, and they did not agree. Resolving the question would require a measurement of the lights as they were operating at the time and evidence showing whether that level of deviation affects detection. Neither is available.

It was assessed as about as likely as not that the lighting affected detection. The matter remains unresolved, and no conclusion is drawn from it.

### **Tailwind**

The possibility that the take-off began with a tailwind above limits was ruled out as very unlikely. The 06:30, 07:00 and 08:00 UTC observations recorded 090 at 7, 6 and 5 kt respectively. The recorded aircraft data showed a headwind component of 4 to 6 kt throughout the roll, with no tailwind component at any point. The crew's performance calculation used a 5 kt headwind, which is consistent with both sources.

### **Fifth sector, 11 hours into duty, 40 minutes behind schedule**

It is virtually certain that the occurrence sector was the fifth of the day. That sector had been added that morning when the rostered crew reported unfit. The departure was 11 hours 00 minutes into a duty that began at 06:30 local, the flight was 40 minutes behind schedule, and the planned duty

of 12 hours 05 minutes was within the limit of 12 hours 30 minutes. Both pilots described themselves as tired but not struggling.

A late sector near the end of a long duty is the kind of condition in which degraded decision-making would be expected. The recorded data, however, did not support that mechanism in this occurrence:

- The power levers reached flight idle 0.6 seconds after the decision to reject.
- Maximum braking was established 2 seconds after the decision, at the fast end of the operator's fleet distribution, against a fleet mean of 3.2 seconds.
- No error of omission occurred during the roll.
- The request for a runway inspection was not a step that any document prompted either pilot to consider.

It was therefore assessed as unlikely that the duty pattern influenced the occurrence.

The exposure is nonetheless a standing one. It is likely to matter to future operations. The evening Tarlton Springs rotation places the last sector within the final hour of a 12-hour duty, at an aerodrome with no company engineering support, no alternate within 200 nm at night and no crewing resilience. A single unfit crew member can turn a four-sector day into a five-sector day on the morning of operation.

## Individual Actions

### Take-off rejected 6 kt above V1

The FDR showed V1 called at 112 kt at 07:30:29. Both power levers were retarded to flight idle from 07:30:32, at 118 kt, three seconds and six knots after V1. The CVR places the first officer's sighting call at 07:30:31, "Roos - roos, roos, on the left, they're on the -". It places the captain's "Stopping. I have control." at 07:30:32. The decision to reject was made and announced in a single exchange, one second after the sighting.

Part B 4.6.4 states: "At or above V1 the take-off shall be continued. The decision to continue is made before V1 is reached." The reject was therefore outside the operator's procedure. The captain said so himself, in interview and in his air safety report. His account was that rotating into two large animals would have left the aircraft airborne at night, with damage that could not be assessed, over country that offered no usable option. Faced with that, he chose the course whose end he could see. It is virtually certain that the take-off was rejected six knots above V1.

The effect of the late start on the stop is clear from the performance figures. The scheduled accelerate-stop distance of 1,780 m assumes that the reject begins at or before V1. It left a margin of 67 m to the distance available. The distance reconstruction placed the aircraft 1,152 m along runway 09 when the reject began, with 695 m of sealed surface remaining. The stop actually achieved from 118 kt took 757 m.

Beginning the stop three seconds and six knots late used up the 67 m margin and more. This was before the loss of No. 2 reverse, discussed under Occurrence Events, is taken into account. It is extremely likely that the late reject contributed to the aircraft not stopping on the sealed surface.

The decision must also be read against the circumstances in which it was made. Section 4.6 contains no guidance on an obstruction seen on the runway at any speed. It offers nothing for a case in which continuing is itself hazardous. The captain had about one second in which to decide. The adequacy of the operator's rejected take-off training, which covers aircraft system failures below V1 only, is argued under Risk Controls. The unresolved question of runway edge lighting intensity is set out under Local Conditions.

This decision followed from the animals being on the runway at all, and from guidance that did not address the situation the crew faced. The investigation therefore regards the rejected take-off above V1 as a contributory factor in the overrun.

### **No runway inspection requested before the first departure after last light**

Neither the CVR nor the tower's recorded audio contains any request for a runway inspection, any offer of one, or any mention of wildlife before departure. The aerodrome's daily inspection log for 18/06/2026 shows one inspection, at 07:35 local before the first arrival, with the movement area found clear. No further inspection was carried out or requested that day.

Aerodrome Manual 6.3 provides for inspections outside the scheduled time at the request of the reporting officer or a pilot in command. The reporting officer confirmed that one would have been carried out if asked. It is virtually certain that no inspection was requested before the departure.

Neither crew member considered asking for one. Nothing in the documents they worked to prompted them to. The captain had expected Tarlton Springs to appear in the operator's list of aerodromes requiring an inspection, and learned only afterwards that it did not. Why that list did not include Tarlton Springs is argued under Risk Controls.

The aerodrome's own wildlife sighting log showed that kangaroos were regularly present on and around the movement area around last light, some of them on the sealed surface. That log was held only as a paper register in the aerodrome office. It was not passed to any operator, and no operator had asked for it. The crew had no route to that knowledge.

An inspection would have covered the section of runway between 1,050 m and 1,250 m, where the fence pad and both carcasses were found. Aerodrome staff disperse animals when they find them. However, an inspection carried out some minutes before the roll cannot guarantee that the runway stays clear afterwards.

On that basis it is likely that the absence of an inspection contributed to the animals being on the runway when the aircraft reached them. It was a contributory factor. It was an omission that no procedure had assigned to anyone, and it is not a lapse by either crew member.

### **Maximum braking and No. 1 reverse applied without delay**

The response to the decision was fast. The power levers reached flight idle 0.6 seconds after the decision. Maximum braking was established at 07:30:34, two seconds after it. This was at the fast end of the operator's recorded fleet distribution, against a fleet mean of 3.2 seconds from decision to maximum braking.

Anti-skid was active on all four wheels. Brake energy reached 84% of the certified maximum without a brake fire. Reverse was selected on No. 1 at 07:30:36. The captain's call at that moment, "Reverse - number two's not coming", reflected the loss of the No. 2 propeller in the strike. The first officer called the speeds and the approaching runway end.

The aircraft left the seal at 21 kt and stopped 62 m into the 90 m RESA. It came to rest upright and on the extended centreline, and the passengers were disembarked without an evacuation. It is virtually certain that this response occurred as the recorded data shows.

Had the stop begun at the fleet mean response, the aircraft would have travelled an estimated further 30 to 40 m. That compares with the 28 m of prepared surface that remained beyond its final position.

It is very likely that the promptness of the braking and reverse application reduced the severity of the outcome. It was the difference between a low-speed overrun into a prepared surface and a

higher-energy one at or beyond the RESA's far edge. The same investigation that finds the reject was outside procedure also finds that what followed was carried out better than the fleet average.

## Occurrence Events

### **Two kangaroos entered the runway strip and moved onto the sealed surface**

The presence of the animals on the runway is not in doubt. Two adult kangaroos were recovered from runway 09, at 1,188 m and 1,221 m from the threshold, both on the northern half. The first officer's call at 07:30:31, "Roos - roos, roos, on the left, they're on the -", places them on the northern side of the aircraft's path. The first officer's account on the flight deck after the stop was consistent with this: "They came out of the dark on the left of the strip, crossed the centreline." The first officer estimated a range of around 200 m at first sight.

The route by which the animals reached the runway is less certain. The site inspection found a clear animal pad through the failed section of the northern fence at the 1,050 m mark. This is 138 m before the first carcass and on the same side of the runway. The positions of the carcasses relative to that pad are the strongest physical indication of the entry point. However, the fence had failed over a continuous 140 m, and the exact crossing point could not be established. For that reason, it was very likely, rather than certain, that the animals entered through the northern fence and moved onto the sealed surface in the manner described.

The event was not an isolated one. The aerodrome's sighting log recorded eleven occasions on which kangaroos were on the sealed surface of runway 09/27. Of the 41 kangaroo entries in the log, 38 fell in the period around last light. No runway inspection had been carried out since 07:35 local that morning.

The deferral of the fence repair, the timing of the departure, and the absence of a runway inspection are each argued in their own sections.

It was extremely likely that the presence of the animals on the runway led directly to both the rejected take-off and the strike. Without them there would have been neither. The animals entering the runway strip and moving onto the sealed surface was therefore a causal factor in the overrun.

### **No. 2 propeller and right main gear struck two kangaroos**

The strike is virtually certain. The recorded data, the physical damage and the recovered animal tissue all agree. The FDR recorded the impact at 07:30:33 at 114 kt, one second after the reject was initiated. It showed a lateral acceleration spike of 0.31 g to the right and a vertical spike of 1.48 g. The CVR recorded a sound consistent with a heavy impact at the same second.

The post-occurrence inspection found the following damage:

- three No. 2 propeller blades damaged beyond repairable limits, one with its tip separated
- animal tissue in two blade roots
- distorted right main gear doors
- abrasion and two punctures in the lower right fuselage skin, consistent with propeller debris.

The significance of the strike lies less in the damage than in its effect on the stop. No. 2 propeller RPM decayed from the moment of impact, and the No. 2 reverse interlock did not make. At 07:30:36 the captain called "Reverse - number two's not coming." Reverse was selected on No. 1 only.

The strike therefore removed half of the aircraft's reverse thrust at the start of a stop that already had no margin. The effect of this on the deceleration is argued below.

It was virtually certain that the strike reduced the stopping performance available. The strike of the No. 2 propeller and right main gear on the two kangaroos was a causal factor in the overrun.

### **Deceleration achieved below the scheduled accelerate-stop performance**

This is the point at which the two lines of the occurrence meet: the late reject and the loss of reverse. It is best understood as a single mechanism.

The scheduled accelerate-stop distance required was 1,780 m, against an ASDA of 1,847 m, leaving a margin of 67 m. That figure assumed a stop begun at or before V1, with symmetric reverse. Neither assumption held on this occasion.

The distance reconstruction from recorded groundspeed placed the aircraft 1,152 m along runway 09 when the reject was initiated. This left 695 m of sealed surface. The stopping distance actually achieved from 118 kt was 757 m. The difference is the 62 m by which the aircraft entered the RESA. This agrees with the nose wheel rut found on the extended centreline and with the continuous tyre marks from 1,160 m to the runway end.

Between 07:30:34 and 07:30:44 the mean deceleration was 2.6 m/s<sup>2</sup>. This was approximately 15% below the value used in the certified accelerate-stop data for this weight and runway condition. The certified data assumes symmetric reverse.

The shortfall was not attributable to the braking system:

- maximum braking was established with anti-skid active on all four wheels
- the anti-skid system tested serviceable afterwards
- the brake energy absorbed was 84% of the certified maximum, with no brake fire or hydraulic leak.

The deficit is consistent with the loss of No. 2 reverse. Once the aircraft left the sealed surface at about 21 kt, deceleration fell to 1.9 m/s<sup>2</sup> in the graded gravel. It came to rest 28 m short of the end of the 90 m RESA.

On the investigation's assessment, both elements were needed to carry the aircraft beyond the runway end, and neither alone would have done so. These were the reject begun 6 kt above V1 and the deceleration approximately 15% below the scheduled value. The reasons for the timing of the reject are argued in the section on individual actions.

It was virtually certain that the deceleration achieved was below the scheduled accelerate-stop performance, and that this shortfall, combined with the late start of the stop, produced the overrun. The shortfall was therefore a causal factor.

### **No. 2 engine power loss during the take-off roll**

A loss of power on the No. 2 engine was considered as a cause of the reject and was ruled out as very unlikely. The evidence for this was as follows:

- both engines held 98.4% and 98.5% torque, with no caution, warning or exceedance, until the power levers were retarded at 07:30:32
- the CVR contains no engine-related call
- the borescope found no pre-existing damage
- the No. 2 RPM decay began only at the moment of impact, after the reject had been initiated.

## Conclusions

The investigation identified the following causal and contributory factors. They should not be read as apportioning blame or liability to any organisation or individual.

## Findings

1. Two of the six floor proximity escape path segments aft of row 16 on the left side of VH-SBQ were dark on battery power because of a failed common ballast. Neither the pre-flight emergency lighting test nor any other check available to the crew or a line engineer could detect this defect. Its existence is virtually certain. It is unlikely that it had any effect on this occurrence or its outcome. It is very likely that it increased the risk to occupants in any evacuation, because the same ballast is fitted across the type and a 24-month functional test interval cannot reveal the fleet's true state. (Safety issue)
2. A tailwind is very unlikely to have existed at any stage of the take-off roll. The aerodrome observations gave a headwind component of 5 to 7 kt and the recorded data gave 4 to 6 kt throughout, consistent with the 5 kt headwind used in the performance calculation.
3. The occurrence took place on the fifth of five sectors, 11 hours 00 minutes into the duty and 40 minutes behind schedule, the fifth sector having been added that morning; this is virtually certain.
  - It is unlikely that this influenced the occurrence. The recorded response to the reject was at the fast end of the fleet's distribution, and no document prompted a runway inspection request, which was the one step the schedule might plausibly have affected.
  - The evening Tarlton Springs rotation places the last sector late in a long duty whenever a crew member reports unfit. It is likely that this pattern is a standing exposure that increases risk.
4. It is virtually certain that the power levers reached flight idle 0.6 seconds after the decision to reject and that maximum braking was established 2 seconds after it, against a fleet mean of 3.2 seconds, with No. 1 reverse selected within 4 seconds. It is very likely that this reduced the severity of the outcome. A stop begun at the fleet mean response would have carried the aircraft an estimated further 30 to 40 m, close to the far edge of the RESA.
5. A loss of power on the No. 2 engine during the take-off roll was very unlikely.
  - Both engines were at matched take-off torque with no caution, warning or exceedance until the power levers were retarded.
  - Borescope examination found no pre-existing damage.
  - The No. 2 propeller RPM decay began at 07:30:33, at the moment of impact and after the reject had begun.

The investigation also examined whether the intensity of the runway edge lighting reduced the crew's ability to see the animals, and this could not be resolved.

- **Evidence for an effect:** A photometric check seven days after the occurrence measured four northern edge lights between the 900 m and 1,200 m markers at 118% to 131% of nominal for stage 3. This was the section the animals crossed. The aerodrome's maintenance programme checked lamp serviceability but never measured intensity. The commander described the lighting as a corridor of light with darkness either side.
- **Evidence against an effect:** The first officer, who saw the animals, described the lighting as normal and saw them clearly once they were on the sealed surface. She acknowledged that, with only two daylight landings to compare against, she was not well placed to

judge. The commander's own non-detection has an explanation that does not depend on the lighting: after V1 his attention was on the engine page, where the procedure places it. The evidence was evenly balanced and no finding is made. Settling the question would require a measurement of the lights as they were on the night and evidence connecting edge light intensity to detection of an object on the runway. Neither exists.

## Causal factors

- The take-off was rejected at 118 kt, six knots and three seconds after V1, outside the operator's procedure requiring the take-off to be continued at or above V1. This is virtually certain. It is very likely that it contributed to the overrun, because the scheduled accelerate-stop distance of 1,780 m assumed a stop begun at or before V1 and left a margin of only 67 m.
- It is very likely that two kangaroos entered the runway strip and moved onto the sealed surface of runway 09. Their presence and final positions were not in doubt, but their exact crossing point through the failed fence was not established. It is very likely that their presence on the runway led directly to both the rejected take-off and the strike.
- With virtual certainty, the No. 2 propeller and right main gear struck two kangaroos at 114 kt, one second after the reject began. The strike virtually certainly contributed to the overrun by destroying the No. 2 propeller and removing reverse on that side, in a stop that had no margin.
- The deceleration achieved, a mean of  $2.6 \text{ m/s}^2$  with reverse on No. 1 only, was virtually certainly about 15% below the value used in the certified accelerate-stop data. It is virtually certain that this shortfall, together with the late start to the stop, carried the aircraft 62 m beyond the runway end. Neither alone would have done so.

## Contributory factors

- It is virtually certain that Aerodrome Standard AS-139 required a documented wildlife hazard management plan only above a size threshold that Tarlton Springs did not meet. The aerodrome was not in breach, but it held no plan, hazard assessment or dispersal programme.
- It is likely that this contributed by leaving the aerodrome's sighting log unconnected to the operators using it and its failed fence on an unfunded works register.
- This judgement is less secure than the other contributory findings. The aerodrome remained obliged to take reasonable measures, and there was no evidence that a documented plan would have secured funding for the fence. (Safety issue)
- It is virtually certain that repair of the northern perimeter fence was deferred, unfunded, at three successive budget rounds after it was recorded as failed in 04/2023 with animals observed passing through.
- It is very likely that the resulting 140 m breach contributed by providing the route by which the kangaroos reached the runway.
- An animal pad through the fence was found at the 1,050 m mark, on the same side as both carcasses, although the exact crossing point was not established. (Safety issue)
- The evening service was very likely approved as a schedule change without reopening route risk assessment RRA-041. The change record did not mention that assessment's daylight-only restriction. The approver could not be interviewed.
- It is likely that this contributed, because a review against a night operation would have reached the aerodrome's wildlife exposure and its absence from Appendix 8A.

- Such a review might instead have been confined to lighting and diversion fuel, the matters the restriction named. (Safety issue)
- Operations Manual Part A 8.14 virtually certainly required a runway inspection before the first departure after last light only at aerodromes listed in Appendix 8A. The sole route onto that list was a reported company wildlife strike, so eleven recorded sightings of kangaroos on the sealed surface at Tarlton Springs could not bring it onto the list.
- It is very likely that this contributed, because an inspection would otherwise have been requested and a clear result required before the roll. (Safety issue)
- It is very likely that the operator's rejected take-off training and Part B 4.6 addressed only aircraft system events and gave no guidance on an obstruction on the runway at any speed. Three of the four training scenarios were below V1.
- It is likely that this contributed to the decision to reject above V1, which was made in the seconds available without any rehearsed frame.
- The commander's own account supports a different reading: that he knew the rule and departed from it deliberately because he judged the alternative worse. On that reading the gap is a fleet-wide deficiency rather than a cause of this reject, and the investigation did not exclude that reading. (Safety issue)
- It is very likely that the take-off roll began in full darkness, 5 minutes after the end of civil twilight with no moon. This was within the period in which the aerodrome's sighting log recorded 38 of its 41 kangaroo entries. It is likely that this contributed, since nothing off the sealed surface could be seen and the animals were not detected until they were on the lit runway.
- It is virtually certain that no runway inspection was requested before the departure. The aerodrome would have carried one out on request, but nothing in the operator's documents prompted either pilot to consider it.
- It is likely that the absence of an inspection contributed, since one would have covered the section where the fence pad and both carcasses were found.
- An inspection some minutes before the roll could not have guaranteed that the runway stayed clear.

## Safety action and recommendations

### Documented wildlife plan required only above a size threshold the aerodrome does not meet

Aerodrome Standard AS-139 requires a documented wildlife hazard management plan only where an aerodrome is used by aeroplanes above 27,000 kg MTOW or serves more than 350,000 passenger movements a year. Tarlton Springs met neither condition. Its heaviest scheduled aeroplane was 22,900 kg, and it handled 61,400 passenger movements in 2025. The aerodrome operator was therefore obliged to take reasonable measures against wildlife but not to document them, and it was not in breach of the standard. The aerodrome held no plan, no hazard assessment and no dispersal programme. Its wildlife measures were undocumented, unaudited and, in practice, unfunded.

The investigation found that this threshold likely contributed to the occurrence. The absence of a documented plan allowed the perimeter fence repair to be deferred for three budget cycles. It also left no route by which the aerodrome's sighting log could reach operators, whose runway inspection requirement applied only at aerodromes listed after a strike. This finding is held with less confidence than the other contributing factors. A plan was not the only means of repairing the fence or sharing the sighting log, and there was no evidence that a plan would have secured funding that the council's budget decisions did not provide. The significance of the issue lies in the threshold itself, which leaves an aerodrome of this size without a documented, owned and audited wildlife programme even when it is fully compliant.

**Safety action taken:** No safety action addressing this issue has been recorded.

**Safety recommendations:** No safety recommendation is made in respect of this issue.

### Perimeter fence repair deferred for three budget cycles

The aerodrome operator deferred repair of the northern perimeter fence at Tarlton Springs for three consecutive budget cycles. It is virtually certain that this deficiency existed. The works register entry WR-221, raised in 04/2023, recorded the following:

- four displaced panels and strained wire between the 700 m and 1,300 m runway markers
- animals observed passing through the fence
- a replacement cost estimated at \$41,000

At the 2023, 2024 and 2025 budget rounds the repair was deferred as unfunded and marked for review at the next cycle. A later entry, WR-263 (02/2026), recorded the same section deteriorating further.

When the investigator inspected the site on 23/06/2026, the failure extended over a continuous 140 m section. The bottom two wires had separated from the posts, and a clear animal pad ran through the fence at the 1,050 m mark. This was on the same side of the runway as the carcasses recovered at 1,188 m and 1,221 m.

The significance of the issue is that the breach was the physical route by which the animals reached the movement area. A fence in that condition does not exclude kangaroos. The deferral very likely contributed to the occurrence by leaving that route open. The condition of the fence and the passage of animals through it had been known and recorded by the aerodrome operator since 2023. The repair was deferred throughout that period, and the fence was not repaired before the occurrence.

No safety action in response to this issue has been recorded. No safety action or recommendation is therefore set out here.

### **Evening service approved as a schedule change, route risk assessment not reopened**

The operator's current route risk assessment for Tarlton Springs, RRA-041 issue 3, restricted operations at the aerodrome to daylight pending review of night lighting and diversion fuel policy. The evening rotation introduced from 14/07/2025 under CMR-2025-118 was extremely likely approved as a schedule change only. The approval did not refer to that restriction and did not reopen RRA-041, and this likely contributed to the occurrence.

The issue matters because of the process, not only the single decision. The change placed a company aircraft at Tarlton Springs after last light for the first time, against a restriction that was still current. It did so through a change assessment that asked whether the change introduced anything new, rather than whether the existing route risk assessment still held.

A restriction set out in a current controlled document could therefore be bypassed without anyone considering it. The Safety Department's involvement went no further than receiving the approved change. Nothing in the process as it stood would prevent the same thing happening with any future change that is characterised as a schedule change only.

No safety action has been recorded in response to this safety issue. No safety recommendation has been recorded in respect of it.

[NOT ESTABLISHED: whether a safety action has been taken or a safety recommendation is to be made in respect of this safety issue]

### **Runway inspection required only at listed aerodromes, and the listing criterion is a strike**

The operator's Operations Manual Part A, section 8.14.2, required the pilot in command to request a runway inspection before the first departure after last light at any aerodrome listed in Appendix 8A, and not to commence the take-off roll until the result was passed. Section 8.14.3 required nothing at other aerodromes. An aerodrome was admitted to Appendix 8A only where a wildlife strike involving a company aircraft had been reported there in the preceding 24 months. No provision in Part A asked crews to report sightings, and no provision allowed an aerodrome to be listed on any other evidence. Nine aerodromes were listed. Tarlton Springs had never been listed, and the list was last reviewed on 03/03/2025, four months before the evening service began.

It is virtually certain that this arrangement existed. It is very likely that it contributed to the occurrence, because it meant that no runway inspection was requested before the departure. Had the requirement applied, a pass would have been required before the roll, and the aerodrome's own staff disperse animals when they find them. However, an inspection completed some minutes before departure does not guarantee a clear runway at the start of the roll.

The safety significance lies in the design of the trigger. The inspection requirement was a sound defence against this kind of occurrence, but the only way to activate it at an aerodrome was a reported strike there, which is the outcome it exists to prevent. Eleven sightings of kangaroos on the sealed surface at Tarlton Springs could not bring the aerodrome onto the list, because none of them was a strike and none was reported to the operator.

No safety action has been recorded in response to this safety issue.

## Rejected take-off training covers aircraft system failures below V1 only

It is extremely likely that the operator's recurrent training for the rejected take-off addressed only aircraft system failures. Module 4 of TRG-SYL-07 contains four scripted scenarios:

- an engine failure at 90 kt
- an engine fire at 100 kt
- a configuration warning at 85 kt
- an engine failure at V1 plus 5 kt, in which the take-off is continued

Three of these arise below V1, and none involves an external obstruction. No crew had been given such a scenario in the five training cycles on file, and the current cycle alone covered 214 crew. Part B 4.6 of the Operations Manual is consistent with the syllabus. It lists reject items to V1 and requires the take-off to be continued at or above V1. It gives no guidance on an obstruction on the runway at any speed.

The significance of the issue extends beyond this occurrence because it applies across the fleet. A crew trained under the current syllabus who meets an event other than an aircraft system failure faces the decision to stop or go without having rehearsed it. It is likely that the absence of such training contributed to the take-off being rejected 6 kt above V1. That finding carries its qualification. The captain stated that he knew the rule and departed from it deliberately. On that reading, the training gap explains the circumstances in which the decision was made rather than the decision itself.

### Safety action taken

No safety action has been recorded in response to this safety issue.

### Safety recommendations

No safety recommendation is made in respect of this safety issue. [NOT ESTABLISHED: whether the operator has taken or proposes any safety action on rejected take-off training for external obstructions]

## Cabin floor proximity emergency lighting partially unserviceable

While walking back through the cabin during the disembarkation, the first officer noticed that sections of the aisle floor strip were unlit. A functional check of the cabin emergency lighting on VH-SBQ on 24 June 2026 found that two of the six floor proximity escape path segments aft of row 16 on the left side were dark on battery power with the system armed. The fault was traced to a failed common ballast.

The pre-flight emergency lighting test on this type illuminates the ceiling lights and exit signs but neither illuminates nor monitors the floor segments. The crew or a line engineer could therefore not have detected the defect without a dedicated functional test. The last such test had been carried out 11 months before the occurrence, within the 24-month programme interval.

It is unlikely that the defect had any bearing on the overrun or its outcome. No evacuation was commanded, the main cabin lights remained on, and the passengers left by the forward airstair in 4 minutes 8 seconds without incident. The defect is nonetheless very likely to be significant. A fire indication, smoke or a night evacuation over the wing could have turned two dark segments in the aft cabin into an obstruction on the escape path. Because the daily test cannot see these segments, the 24-month interval leaves the true state of the fleet unknown, and the same ballast is fitted across the type.

No safety action has been recorded in response to this safety issue, and no safety recommendation is made.

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